

PDSDA Capstone — Project Schedule

Posture Detection System for Smart Desk Automation · MEng Robotics & Intelligent Autonomous Systems · University of Cincinnati · May 11 – July 22, 2026

Wk	Dates	Phase	Deliverables / Key Activities	Status
1	May 11–17	Planning	Project scoping · advisor kick-off · technology evaluation · V-Model framework adopted	Complete
2	May 18–24	Requirements	SRS v1.0 drafted · WP1.2 Evaluation Framework · Dataset Collection Protocol · CR-01 to CR-06	Complete
3	May 25–31	Requirements	SRS v1.0 baselined · WP1.1 Change Log · WP2.1 Literature Review v3.0 · Capstone Proposal v2	Complete
4	Jun 2–8	Architecture	WP2.2 System Architecture · WP2.3 Fault Tree Analysis (4 MCS) · WP3.0 FMEA (29 failure modes) · Dataset generation pipeline	Complete
5	Jun 9–15	Implementation	WP5.0 Validation Plan · generate_dataset.py · 754 frames collected (417 dev + 337 test) · WP2.2 Data Collection Log · CR-07–09	Complete
6	Jun 16–22	V&V	Claude API: 99.4% acc, 2,681 ms (NFR-003 ✓) · Gemma 4 E4B: 96.4% acc, 5,593 ms (NFR-002 ✗ latency) · Label review: 2 confirmed bad labels · Drive populated	Complete
7	Jun 23–29	V&V	Gemma 4 26B: 98.5% acc, 1,855 ms (NFR-001 ✓) · T1–T8 reliability all PASS · VAL-01–04 UAT all Accept · Final Report v6 · Code rev14 (CAM-01/02/03)	Complete
8	Jun 30–Jul 6	Report	Final Report polish · presentation deck draft · advisor review by Jul 14	Upcoming
9	Jul 7–13	Presentation	Presentation deck finalised · rehearsal · CEAS portal forms submitted · advisor sign-off	Upcoming
10	Jul 14–22	Submission	Final report submitted · presentation delivered · CEAS forms filed · project archived	Jul 22

Key Milestones

May 31	SRS v1.0 baselined — all downstream documents trace to this baseline
Jun 21	Dataset complete: 754 frames (exceeds 270-frame CR-09 minimum by 2.8x)
Jun 24	All three approaches benchmarked — Claude 99.4%, Gemma 4 26B 98.5%, Gemma 4 E4B 96.4%
Jun 27	Full reliability suite T1–T8 and UAT VAL-01–04 complete — all PASS / Accept
Jun 27	Final Report v6 and code rev14 delivered — all sections and bugs resolved
Jul 14	Advisor review deadline
Jul 22	Final submission — report, presentation, CEAS portal forms